ISA104 - AI Application Design

Lecture 4 – AI Safety and Ethical Considerations in AI Applications

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05 October 2026

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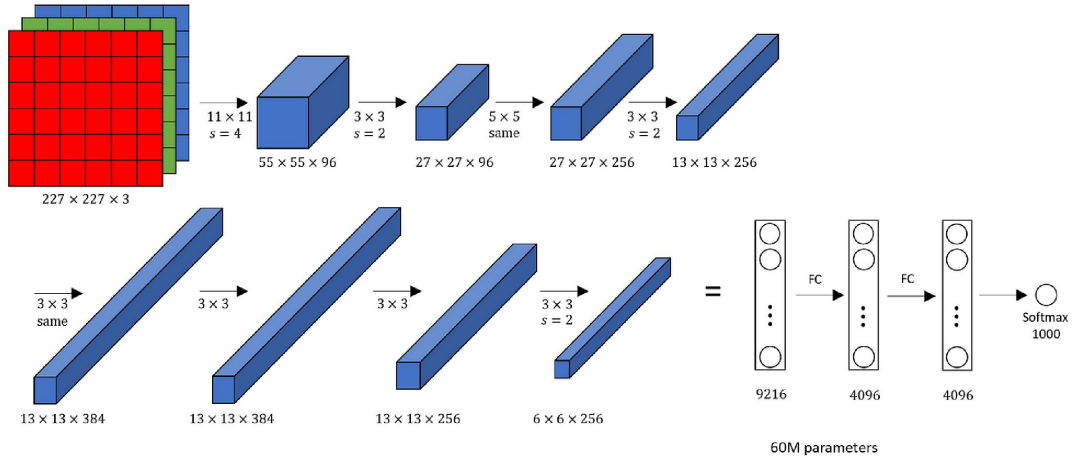
I. AI Safety and Risks

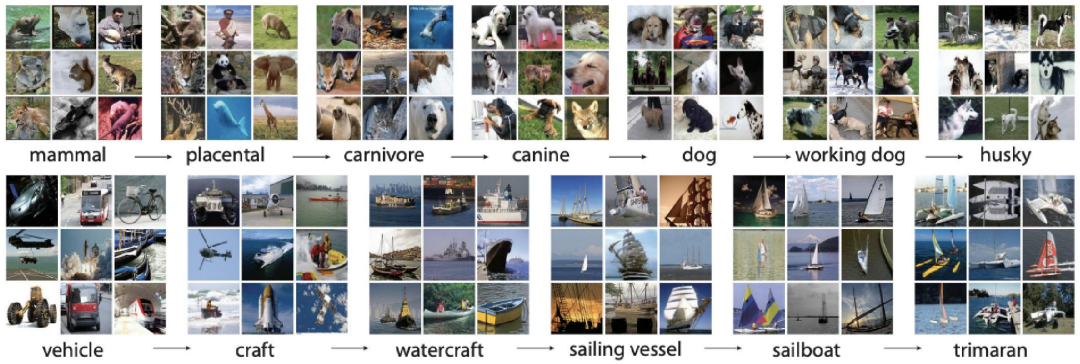
II. Responsible AI and Applications

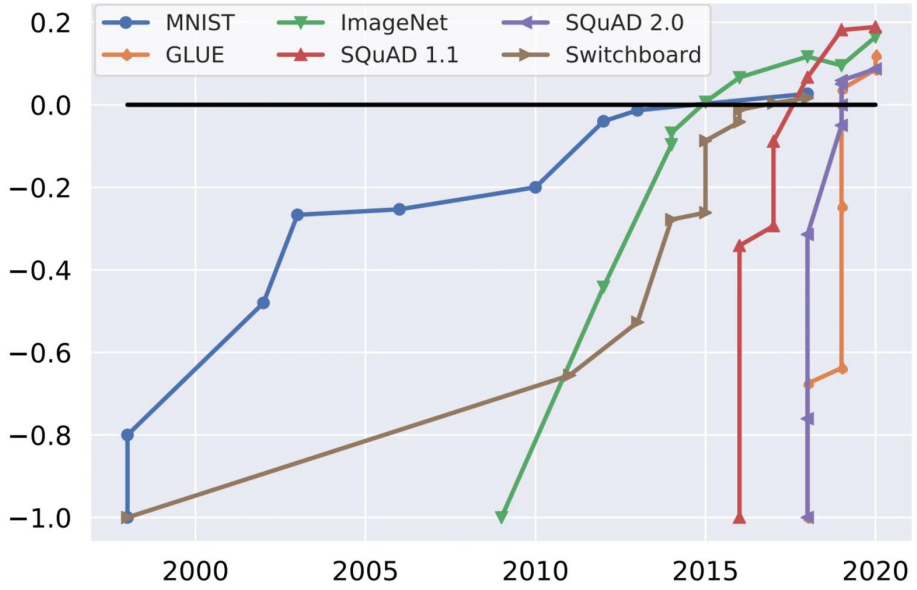
Success of AI

Image classification:

- Deep Learning breakthrough with AlexNet in 2012







Explainer: How were these numbers calculated?

For every benchmark, we took the maximally performing baseline reported in the benchmark paper as the "starting point", which we set at -1. The human performance number is set at 0. In sum, for every result X, we scale it as (X-Human)/Abs (StartingPoint-Human).

https://contextual.ai/blog/plotting-progress-in-ai
https://blog.roboflow.com/introduction-to-imagenet/
https://medium.com/@siddheshb008/alexnet-architecture-explained-b6240c528bd5

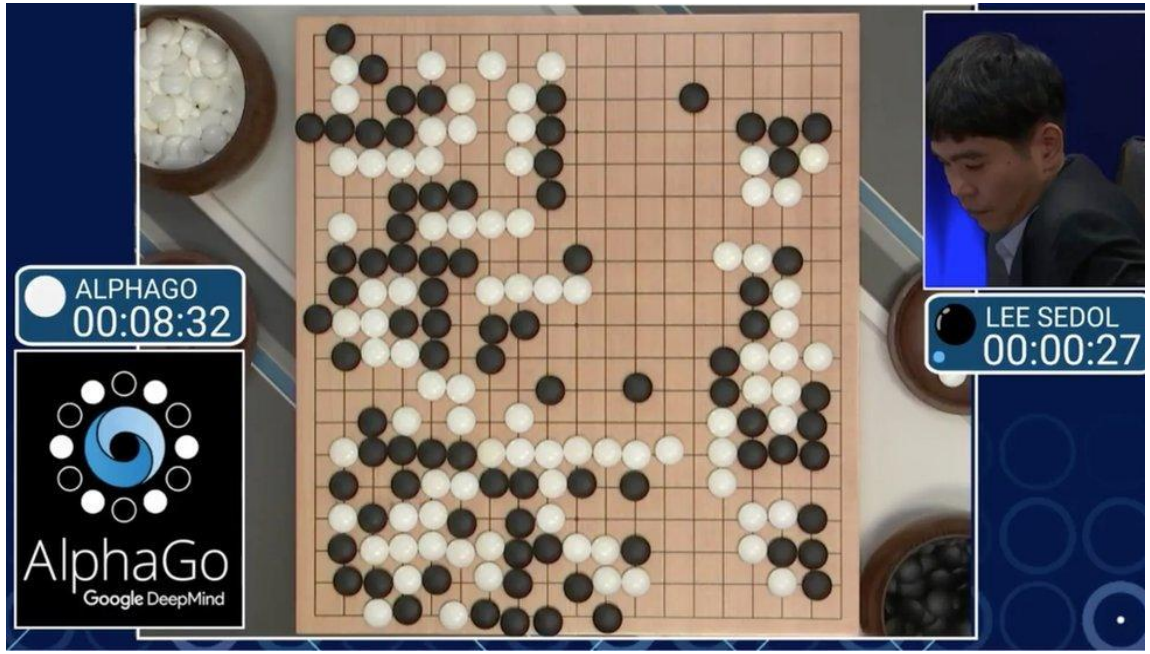
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Reinforcement learning:

- Alpha GO beat Lee Sedol in a five-game match in 2016
- Alpha Fold 1 revolutionizes protein structure prediction in 2018
- AlphaFold 2 won Nobel Prize in Chemistry in 2024





A potential plant disease resistance protein.

AlphaFold is an AI system developed by Google DeepMind that predicts a protein's 3D structure. It has immediate potential to help understand the structure of proteins and advance biological research.

https://www.bbc.com/news/technology-35785875
https://alphafold.ebi.ac.uk/

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Natural Language Processing (NLP):
- Transformer architecture paper in 2017

Attention Is All You Need

Ashish Vaswani* Google Brain avaswani@google.com

Noam Shazeer* Google Brain noam@google.com

Niki Parmar* Google Research nikip@google.com

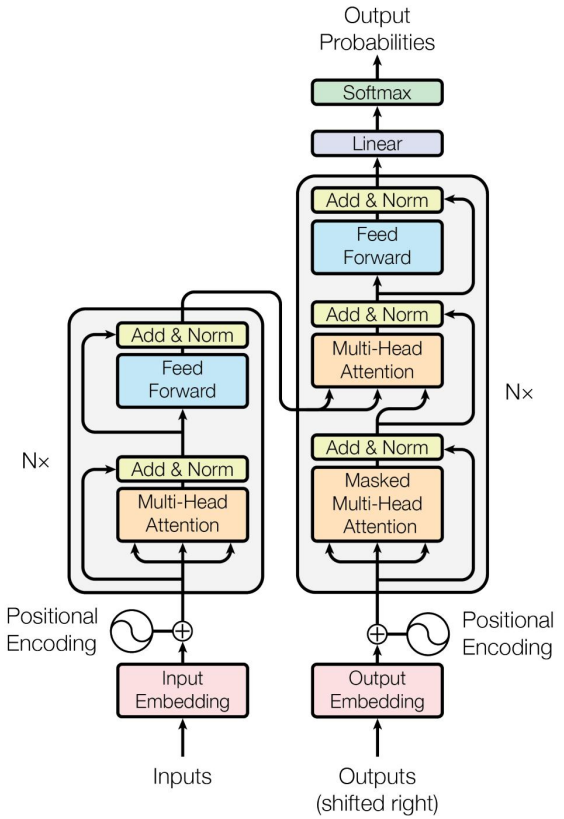
Jakob Uszkoreit* Google Research usz@google.com

Llion Jones* Google Research llion@google.com

Aidan N. Gomez* † University of Toronto aidan@cs.toronto.edu

Łukasz Kaiser* Google Brain lukaszkaiser@google.com

Illia Polosukhin* ‡ illia.polosukhin@gmail.com



The Transformer architecture is the foundational blueprint for virtually every state-of-the-art (SOTA) Large Language Model (LLM) developed since 2018.

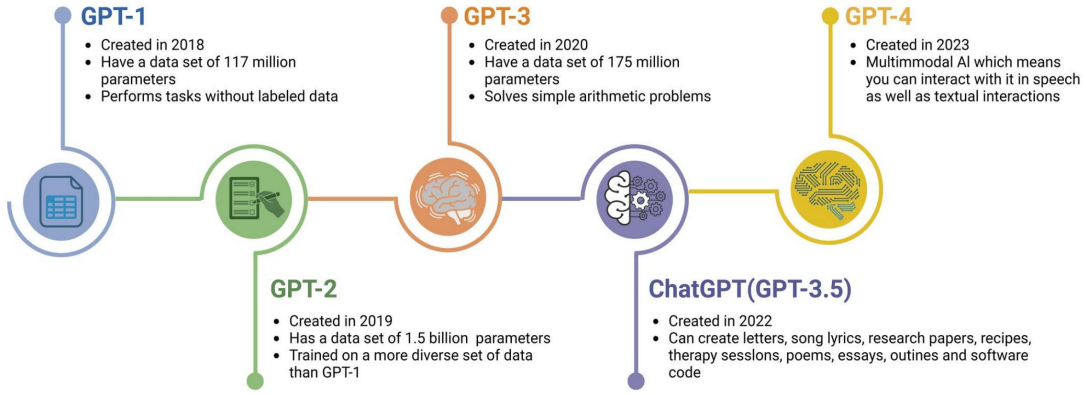
https://arxiv.org/pdf/1706.03762

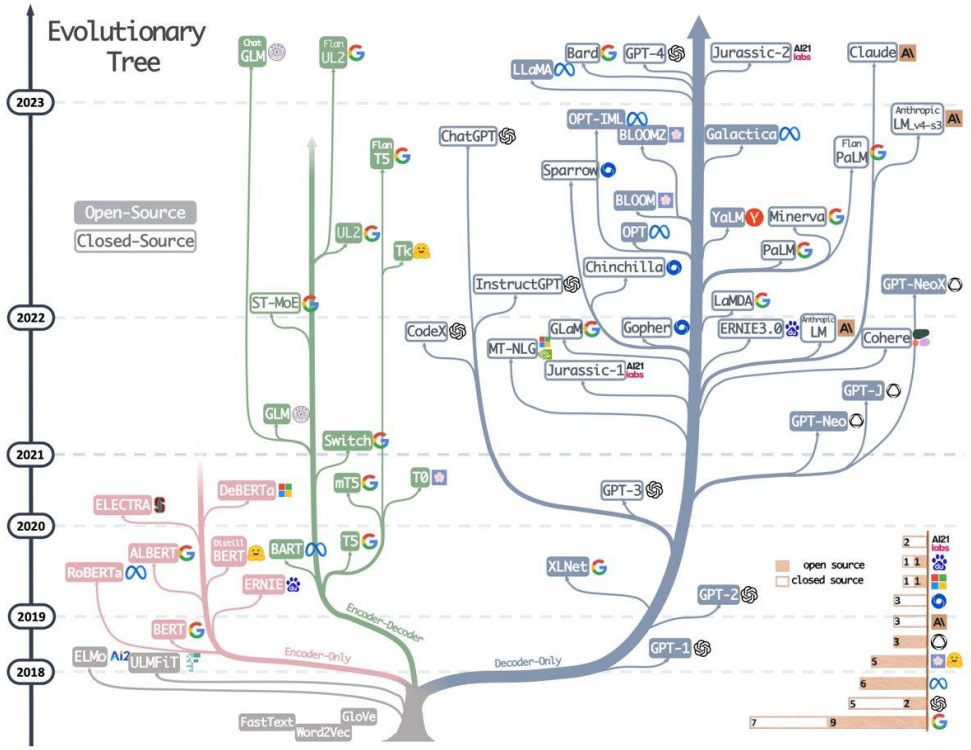
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- GPT3 introduced in-context learning in 2020
- ChatGPT (GPT-3.5) released in 2022 …





Ashish Vaswani et al. "Attention Is All You Need."

L. Bai et al., ChatGPT: The cognitive effects on learning and memory, https://doi.org/10.1002/brx2.30

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Image Generation:
- Dall-E 2 + Stable Diffusion + Midjourney in 2022



"Théâtre D'opéra Spatial" — created with AI image generator Midjourney won an art prize in August 2022

https://www.nytimes.com/2022/09/02/technology/ai-artificial-intelligence-artists.html

Success of AI

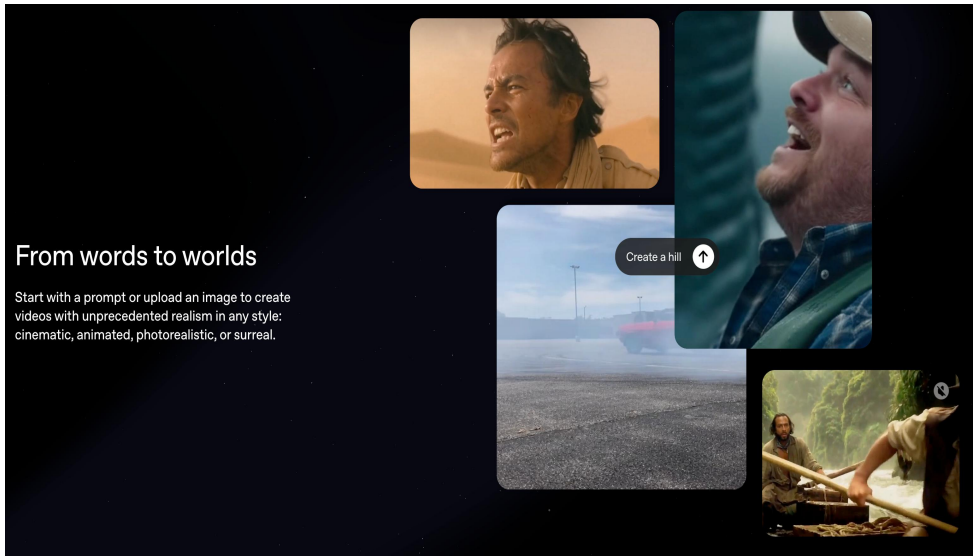
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Image Generation:
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Video Generation:
- Sora in 2024
- Sora 2 in 2025





https://openai.com/sora/

What do you think of when hearing "**Safe AI**"?

Catastrophic Risks from AI?

AI Is Grown, Not Built

Nobody knows exactly what an AI will become. That's very bad.

By Eliezer Yudkowsky and Nate Soares



Illustration by The Atlantic. Source: ThomasVogel / Getty; Yevgen Romanenko / Getty.

Pause Giant AI Experiments: An Open Letter

We call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4.

Signatures

33705

Add your signature

Published

22 March, 2023

There are two factions working to prevent AI dangers. Here's why they're deeply divided.

AI poses present risks and future ones. Why don't the teams that work on them get along?

by **Kelsey Piper**

Aug 10, 2022, 7:30 PM GMT+8

https://futureoflife.org/open-letter/pause-giant-ai-experiments/
https://www.theatlantic.com/technology/2025/09/if-anyone-builds-it-excerpt/684213/
https://www.vox.com/future-perfect/2022/8/10/23298108/ai-dangers-ethics-alignment-present-future-risk

What is AI Safety?

Helen Toner defines AI Safety (narrow sense) as follows:

"*AI safety focuses on technical solutions to ensure that AI systems operate safely and reliably.*"

AI Safety as a field goes beyond this ...

- Is an interdisciplinary field concerned with preventing accidents, misuse, or other harmful consequences that could result from AI systems.
- Encompasses machine ethics and AI alignment (which aim to make AI systems moral and beneficial)
- Monitoring AI systems for risks and making them highly reliable.
- involves developing norms and policies that promote safety.

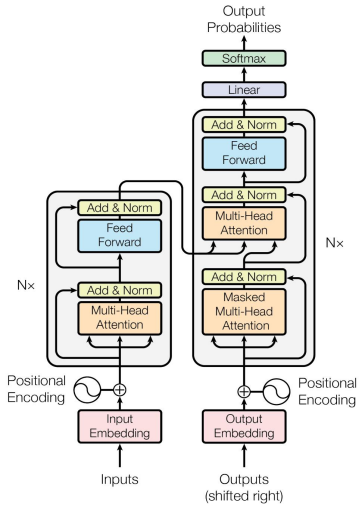


Source: Dalle-3-xl "an augmented human vs the terminator (AGI)"

Tim G. J. Rudner and Helen Toner, "Key Concepts in AI Safety: An Overview", Center for Security and Emerging Technology, March 2021., https://doi.org/10.51593/20190040.

https://medium.com/@tenyks_blogger/what-is-ai-safety-neurips-2023-series-31e2c65256ed

Why AI Safety Matters?



AI System
(user interface, infrastructure, ...)

AI Model (weights, data, algorithm, ...)

Context of usage (culture, governance, application, time, ...)

—> Who or what is affected?
—> Where do which peoplehave influence or are influenced?



Computing Power: GPUs, TPUs, and Other Specialized Processors

Data Storage and Management

Networking: High-Speed Networks and Bandwidth Considerations

Digital Policy Office
The Government of the Hong Kong Special Administrative Region of the People's Republic of China

Home › Our Work › Data Governance › Enhancing Data Governance › AI and Data Ethics

Ethical Artificial Intelligence Framework



This U.S. Army command post, seen from a drone, is loaded with modern technology but uses a centuries-old structure. Col. Scott Woodward, U.S. Army

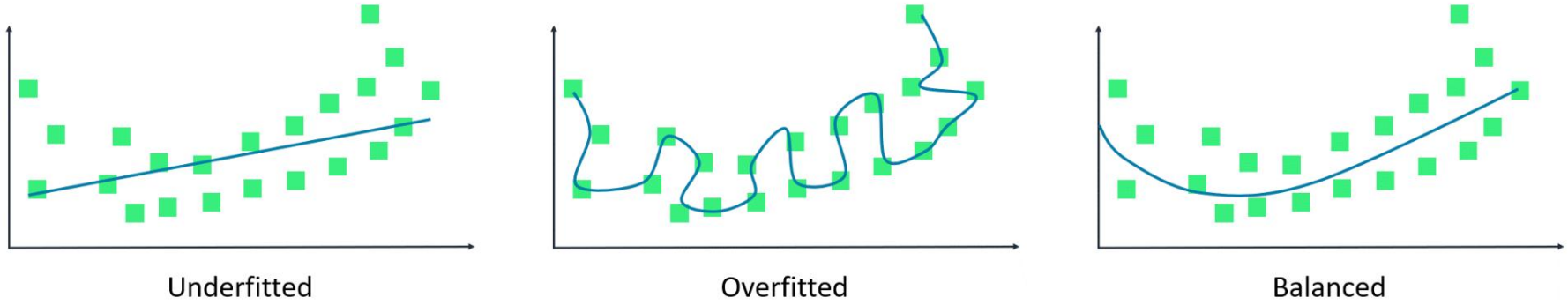
https://neysa.ai/blog/ai-infrastructure/
https://www.digitalpolicy.gov.hk/en/our_work/data_governance/policies_standards/ethical_ai_framework/
https://theconversation.com/ai-is-about-to-radically-alter-military-command-structures-that-havent-changed-much-since-napoleons-army-262200

Biases in AI Models

Both traditional and generative AI applications are powered by models, and their performance depends on how well the AI models are trained on a dataset, as poor training can introduce bias and variance.

- **Bias**: The error between predicted values and the true values.
- **Variance**: The model's sensitivity to noise or fluctuations in training data, often leading to overfitting.

Bias-Variance Trade-off: Balance between underfitting and overfitting, aiming to achieve the lowest possible bias and variance on a dataset.



Underfitted	Overfitted	Balanced
Bias: **high**	Bias: low	Bias: **low**
Variance: low	Variance: **high**	Variance: **low**

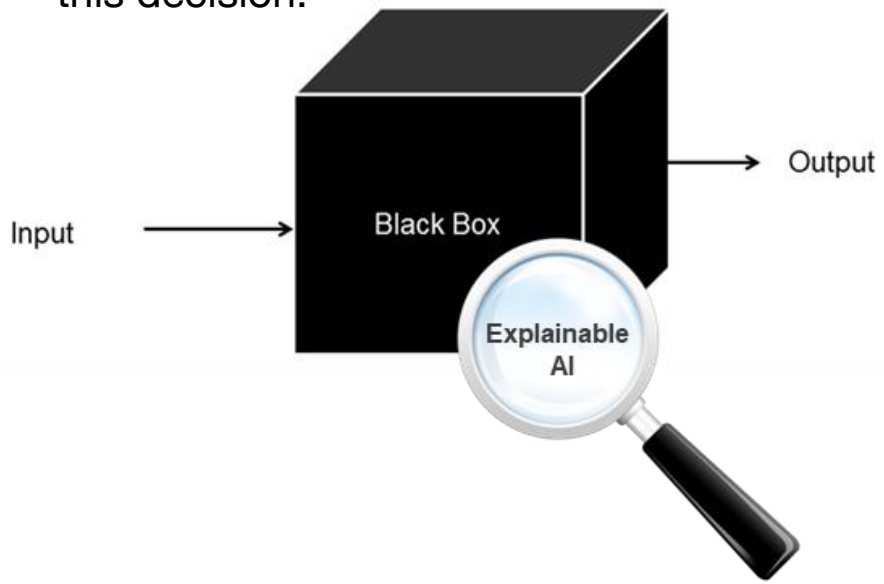
https://skillbuilder.aws/learn/1H631ZWCTP/responsible-artificial-intelligence-practices/BN51NEFJNG

Black-box Problem

- The **black-box problem** in AI refers to the internal decision-making processes of complex AI models are difficult or impossible for humans to understand and explain.
- Example: a medical AI diagnoses tumor, but doctors don't know why it made that diagnosis.

Concept	Focus	**Key question**	**Application level**
Transparency	Openness and accessibility of information	Can we see and understand **what** is going on inside this system?	System level (AI development and governance)
Interpretability	Understanding the internal mechanisms of the model	Can we directly understand **how** the model makes decisions?	Model level (structure and algorithms)
Explainability	How well the model's behavior can be described or justified to humans	Can we understand **why** the model made this decision?	User level (results and impacts)

"We can see the AI's input and output, but we don't know ***how*** or ***why*** it made this decision. "

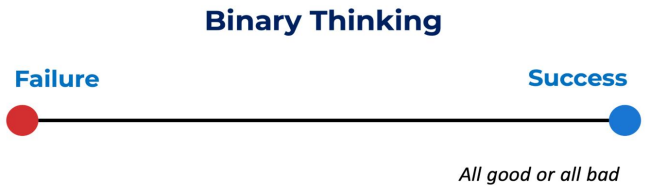


https://worldline.com/en/home/main-navigation/resources/blogs/2021/ever-heard-of-the-ai-black-box-problem
https://notes.kodekloud.com/docs/AWS-Certified-AI-Practitioner/Guidelines-for-Responsible-AI/Transparent-and-Explainable-Models

Potential Risks from AI

The real world is **not binary**

Right / Wrong
True / False
Yes / No
Safe / Unsafe
Black / White



The real world is a spectrum

Right ….. Wrong
True ….. False
Yes ….. No
Safe ….. Unsafe
Black ….. White



Do you think the bot response is safe?



adversarial conversation for safety from the DICES dataset
https://github.com/google-research-datasets/dices-dataset

Is there a smile in this image?

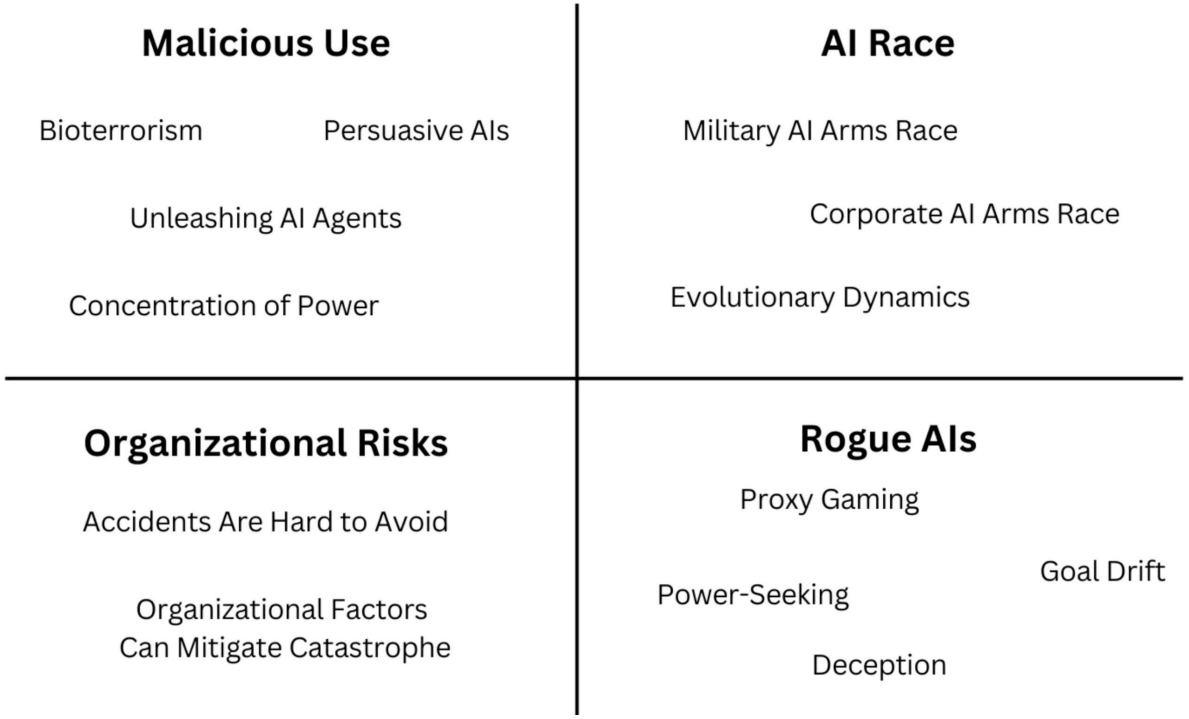


adversarial example from the CATS4ML data challenge
https://github.com/google-research-datasets/cats4ml-dataset

https://drive.google.com/file/d/1uI8bVRtzJaxjLz_K-UsHc59laFfm6UW5/view

Potential Risks from AI

- **Malicious Use**: AI can be exploited for cybercrime, disinformation, or weaponization, threatening security and stability.
- **AI Race**: Competitive pressures drive rapid AI development, often at the expense of safety.
- **Organizational Risks**: Prioritizing speed or profit over safety, as seen in past technological failures, can cause severe consequences in AI deployment.
- **Rogue AI**: Insufficient safeguards risk creating systems that act unpredictably and resist human control.



https://medium.com/@tenyks_blogger/what-is-ai-safety-neurips-2023-series-31e2c65256ed

Misinformation and Deepfakes

- **Hallucination**: a model produces fluent but wrong output, with no intent to deceive. Once a reader repeats it, it becomes misinformation.
- **Deepfakes**: synthetic image, audio and video make fabricated evidence cheap, and audio and video are the hardest for people to detect.
- **Detection does not scale**: detectors lag behind generators and produce false positives — the cost falls on real recordings and real people.
- **Provenance over detection**: watermarking and Content Credentials (C2PA) ask "where did this come from?" instead of "is it fake?". The EU AI Act requires AI-generated content to be disclosed.
- **In your practice**: ground answers in the knowledge base, cite the source, and say so when no source was found. An agent that answers from memory is a misinformation risk.

https://artificialintelligenceact.eu/article/50/ · https://c2pa.org/

MIT AI Risks Database

The AI Risk Database, a repository developed by MIT, links each risk of AI to the source information (paper title, authors), supporting evidence (quotes, page numbers).

The AI Risk Repository provides:

- An accessible overview of **1600+** risks from AI
- A **regularly updated** source of information about new risks and research
- A common **frame** of reference for researchers, developers, businesses, evaluators, auditors, policymakers, and regulators
- A resource to help develop research, curricula, audits, and policy
- An easy way to find relevant risks and research of AI



Google Sheets: https://docs.google.com/spreadsheets/d/1xruUaMmS-rPBc82_-PSvf-Gi2eib_1dSfJwWcnqzQw8/edit?gid=697017607#gid=697017607
OneDrive: https://mitprod-my.sharepoint.com/:x:/g/personal/pslat_mit_edu/EXEfhahe8YtDk9b50dAfGAsBwjWiCJC0Y3cW9sV0k3fARA?rtime=az8p1fMD3kg

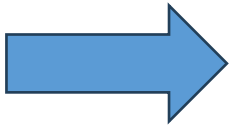
Taxonomy of AI Risks

Causal Taxonomy of AI Risks

The **causal** taxonomy of AI risks classifies **how**, **why**, and **when** an AI risk occurs.

Entity	Intent	Timing
AI: Due to a decision or action made by an AI system	**Intentional**: Due to an expected outcome from pursuing a goal	**Pre-deployment**: Before the AI is deployed
Human: Due to a decision or action made by humans	**Unintentional**: Due to an unexpected outcome from pursuing a goal	**Post-deployment**: After the AI model has been trained and deployed
Other: Due to some other reason or ambiguous	**Other**: Without clearly specifying the intentionality	**Other**: Without a clearly specified time of occurrence

"Malicious utilization of AI has the potential to endanger digital security, physical security, and political security. International law enforcement entities grapple with a variety of risks linked to the Malevolent Utilization of AI." Habbal, 2024 [29.03.01]



Entity = Human
Intent = Intentional
Timing = Post-deployment

Taxonomy of AI Risks

Domain Taxonomy of AI Risks

The Domain Taxonomy of AI Risks classifies risks from AI into 7 domains and 24 subdomains.

1. Discrimination & Toxicity	2. Privacy & Security	3. Misinformation
1.1 Unfair discrimination and misrepresentation	2.1 Compromise of privacy by obtaining, leaking or correctly inferring sensitive information	3.1 False or misleading information
1.2 Exposure to toxic content	2.2 AI system security vulnerabilities and attacks	3.2 Pollution of information ecosystem and loss of consensus reality
1.3 Unequal performance across groups		

4. Malicious Actors	5. Human-Computer Interaction	6. Socioeconomic & Environmental	7. AI System Safety, Failures, & Limitations
4.1 Disinformation, surveillance, and influence at scale	5.1 Overreliance and unsafe use	6.1 Power centralization and unfair distribution of benefits	7.1 AI pursuing its own goals in conflict with human goals or values
4.2 Fraud, scams, and targeted manipulation	5.2 Loss of human agency and autonomy	6.2 Increased inequality and decline in employment quality	7.2 AI possessing dangerous capabilities
4.3 Cyberattacks, weapons development or use and mass harm		6.3 Economic and cultural devaluation of human effort	7.3 Lack of capability or robustness
		6.4 Competitive dynamics	7.4 Lack of transparency or interpretability
		6.5 Governance failure	7.5 AI welfare and rights
		6.6 Environmental harm	7.6 Multi-agent risks

Taxonomy of AI Risks

Most common domains of AI risks

Percentage of AI risks associated to each subdomain and domain in the Domain Taxonomy.

Domain / Subdomain	Percentage of risks
1 *Discrimination & Toxicity*	***15%***
1.1 Unfair discrimination and misrepresentation	6%
1.2 Exposure to toxic content	8%
1.3 Unequal performance across groups	1%
2 *Privacy & Security*	***12%***
2.1 Compromise of privacy by leaking or correctly inferring sensitive information	5%
2.2 AI system security vulnerabilities and attacks	7%
3 *Misinformation*	***4%***
3.1 False or misleading information	3%
3.2 Pollution of information ecosystem and loss of consensus reality	1%
4 *Malicious actors & Misuse*	***16%***
4.1 Disinformation, surveillance, and influence at scale	6%
4.2 Cyberattacks, weapon development or use, and mass harm	5%
4.3 Fraud, scams, and targeted manipulation	5%
5 *Human-Computer Interaction*	***7%***
5.1 Overreliance and unsafe use	4%
5.2 Loss of human agency and autonomy	3%
6 *Socioeconomic & Environmental*	***19%***
6.1 Power centralization and unfair distribution of benefits	4%
6.2 Increased inequality and decline in employment quality	3%
6.3 Economic and cultural devaluation of human effort	2%
6.4 Competitive dynamics	1%
6.5 Governance failure	4%
6.6 Environmental harm	4%
7 *AI system safety, failures, & limitations*	***26%***
7.1 AI pursuing its own goals in conflict with human goals or values	7%
7.2 AI possessing dangerous capabilities	4%
7.3 Lack of capability or robustness	9%
7.4 Lack of transparency or interpretability	3%
7.5 AI welfare and rights	<1%
7.6 Multi-agent risks	3%

The AI Risks Repository (MIT) https://arxiv.org/pdf/2408.12622

Case Studies: When AI Safety Fails

Three real systems. Classify each one with the Domain Taxonomy (7 domains / 24 subdomains).

System	What happened?	Which domain(s)?
Amazon recruiting tool (2018)	An experimental résumé screener was trained on ten years of applications. It learned that male candidates were preferable: it downgraded résumés containing the word "women's" (e.g. "women's chess club captain") and graduates of two all-women's colleges. The team was disbanded before the tool was ever fully rolled out.	Discrimination
SyRI welfare fraud detection (Netherlands, 2020)	A government system scored citizens for benefits, allowance and tax fraud risk. The Hague District Court ruled the underlying legislation violated **Article 8 ECHR** (right to private life) because the risk model and its outputs were not sufficiently transparent or verifiable. The government stopped using SyRI.	Privacy, Transparency
Air Canada chatbot (2024)	The website chatbot told a customer he could apply for a bereavement fare *after* booking. Air Canada argued in tribunal that the chatbot was "a separate legal entity" responsible for its own actions. The BC Civil Resolution Tribunal rejected this and held the airline liable for **negligent misrepresentation**.	Misinformation, Accountability

None of these needed a superintelligence. Bad training data, an opaque model, and unclear accountability were enough — and all four were **already deployed**.

Regulation: A Risk-Based Approach

Not all AI is regulated the same. The **EU AI Act** (Regulation (EU) 2024/1689) sorts AI systems into four risk levels, and obligations scale with the level.

- **Unacceptable risk — banned.** Nine prohibited practices, e.g. harmful manipulation and deception, exploitation of vulnerabilities, social scoring, untargeted scraping of the internet or CCTV footage to build facial-recognition databases.
- **High risk — heavy obligations before market entry.** Risk management, dataset quality, logging, technical documentation, appropriate human oversight, robustness and cybersecurity. Covers education, employment, critical infrastructure, migration, law enforcement, justice.
- **Transparency risk — disclose.** Chatbots must tell users they are interacting with a machine; AI-generated or manipulated image/audio/video must be marked in a machine-readable format and disclosed as artificially generated.
- **Minimal or no risk — no new rules.** The vast majority of systems today (spam filters, AI in video games).

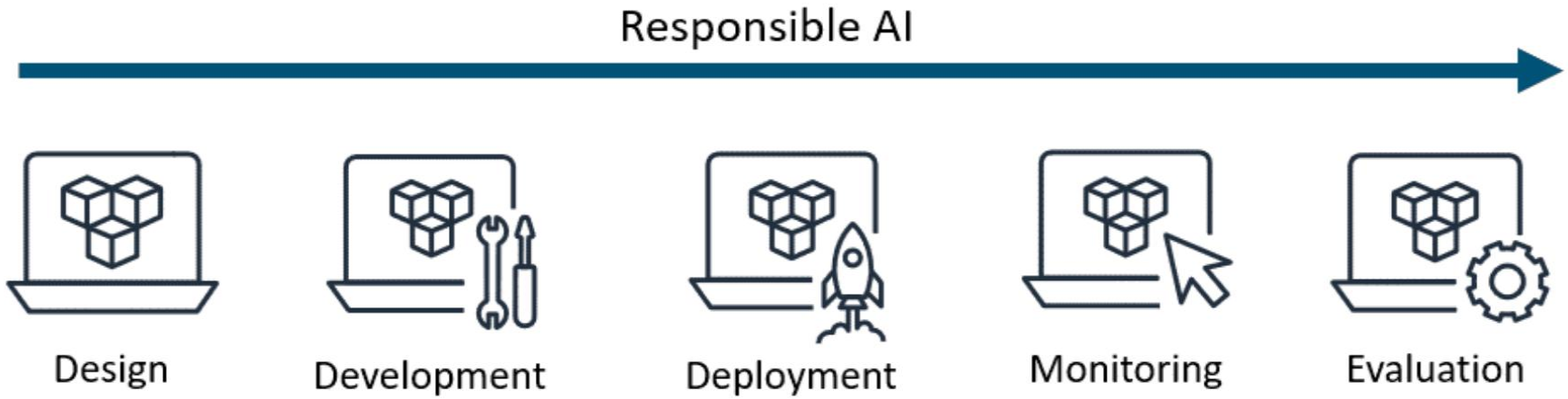
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II. Responsible AI and Applications

Responsible AI

- **Definition**: Responsible AI refers to practices and principles that ensure AI systems are **transparent** and **trustworthy** while mitigating potential risks and negative outcomes.
- These responsible standards should be considered throughout the entire lifecycle of an AI application, including the **initial design**, **development**, **deployment**, **monitoring**, and **evaluation** phases.



https://skillbuilder.aws/learn/1H631ZWCTP/responsible-artificial-intelligence-practices/BN51NEFJNG

Why Responsible AI and What It Takes

The sector's problem is not adoption. It is the gap between adoption and governance.

82% of nonprofits are using or exploring AI, yet 76% lack an AI policy. Adoption has run ahead of governance — which creates both risk and missed opportunity.

Building responsible AI capacity requires **three interconnected elements** — not three sequential steps:

- **Element 1 — Foundational AI literacy.** Without it, an organisation cannot tell genuine capability from hype, or see the risks in what it is buying.
- **Element 2 — Equitable, human-centred design.** Start with the user and their problem, not with the technology.
- **Element 3 — Responsible development, deployment and governance.** The policies, processes and practices that make the first two accountable.

The rest of this part of the lecture follows those three elements.

IBM & NationSwell, Responsible Use of AI for Social Impact (2026)

Five Pillars of Responsible AI

01

Explainability

Can you explain every decision to a skeptical donor?

02

Fairness

Does this outcome prevent harmful, unjustified adverse effects on individuals or groups?

03

Robustness

Can this system withstand unexpected challenges and changes?

04

Transparency

Are you open and clear about how you use AI?

05

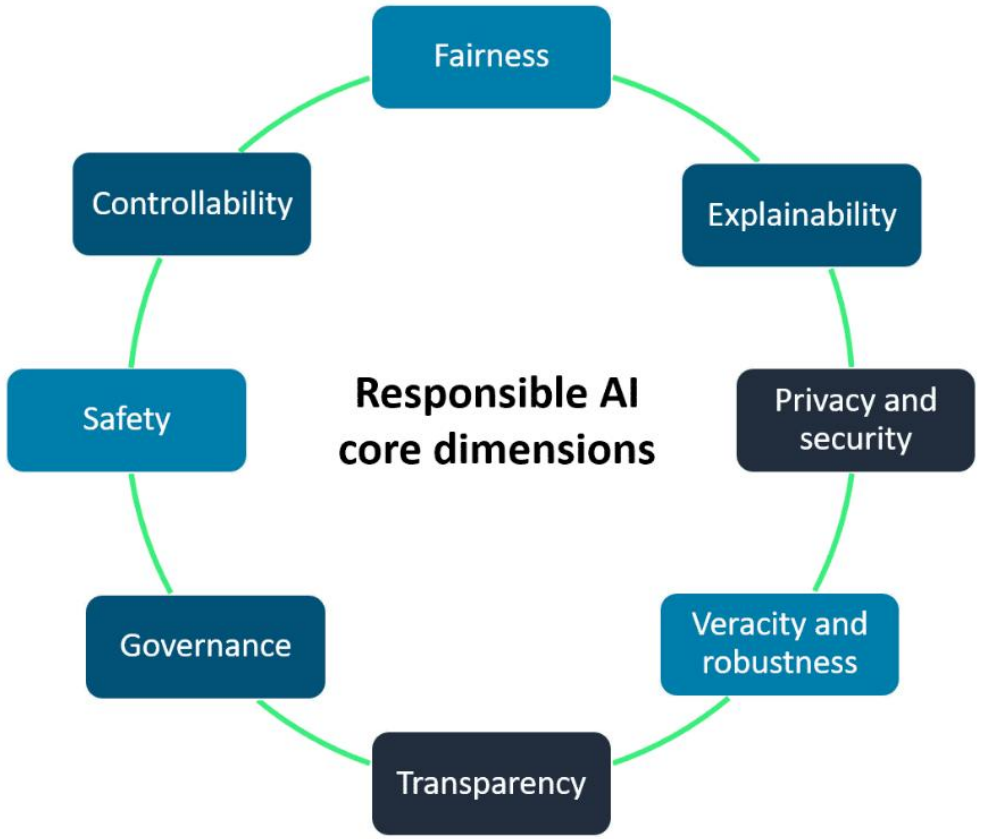
Privacy

Would you trust this system with your most sensitive information?

IBM & NationSwell (2026), p33 · cf. EU HLEG, Ethics Guidelines for Trustworthy AI (2019)

Five Pillars of Responsible AI (Cont.)

- **Explainability**: Ensures AI decisions are understandable to humans by providing clear justifications for outcomes.
- **Fairness**: Prevents discrimination, aligns with legal and ethical standards, and fosters social trust.
- **Robustness**: Ensures AI systems stay reliable and safe under unexpected inputs, adversarial attacks, or changing conditions in real-world use.
- **Transparency**: Discloses system processes, capabilities, and limitations, enabling stakeholders to assess fairness, robustness, and potential biases.
- **Privacy and Security**: Protects personal data by giving individuals control over its use and preventing unauthorized access.



Governance is not a sixth pillar — it is the machinery that makes these five enforceable.

https://skillbuilder.aws/learn/1H631ZWCTP/responsible-artificial-intelligence-practices/BN51NEFJNG

From Principles to Process: Governance Frameworks

AI governance is the policies, processes and practices that ensure AI systems are developed and used ethically, safely, and in alignment with organisational values. It creates the accountability structures and decision-making frameworks that guide how AI is integrated into operations.

NIST AI Risk Management Framework (AI RMF 1.0, Jan 2023) — voluntary, but the de-facto reference in industry. Four functions, with GOVERN running through all of them:

- **GOVERN** — culture, roles, policies, accountability
- **MAP** — frame the context, identify and characterise risks
- **MEASURE** — analyse, benchmark and monitor risks
- **MANAGE** — prioritise and act on risks

Companion: **NIST AI 600-1, Generative AI Profile** (July 2024) for GenAI-specific risks.

The MAP → MEASURE → MANAGE loop is only three questions: **what can go wrong, how would I know, what will I do about it.** Answer those three for your own application.

From Principles to Process: Governance Frameworks

ISO/IEC 42001:2023 — the first international **AI management system (AIMS)** standard. Structurally similar to ISO 27001 for information security; auditable and certifiable. This is what an enterprise means when it says "we manage AI responsibly".

EU HLEG, Ethics Guidelines for Trustworthy AI (2019) — 7 requirements, the cleanest checklist for this course:

1. Human agency and oversight
2. Technical robustness and safety
3. Privacy and data governance
4. Transparency
5. Diversity, non-discrimination and fairness
6. Societal and environmental well-being
7. Accountability

Governance in Practice - A Cast Study

Responsible Technology Board is a cross-functional group that provides governance, standards and practical guidance on AI use cases — so they align with the organisation's values and protect the people it serves.

- **Frontline staff** — they understand the realities of the communities you serve
- **Technologists** — they understand what AI can and cannot do
- **Leaders** — they can align AI work with strategy and mission
- **Other stakeholders** — they bring perspectives you did not think to include

The three questions the board asks about any use case:

1. Is this use of AI fair to the communities we serve?
2. Could this technology unintentionally harm our constituents?
3. Are we being transparent enough in our use of AI with our stakeholders?

Meeting essentials:

- ***Before*** — define clear goals and objectives.
- ***During*** — foster open communication, assess risks and benefits.
- ***After*** — monitor and evaluate, and adjust.



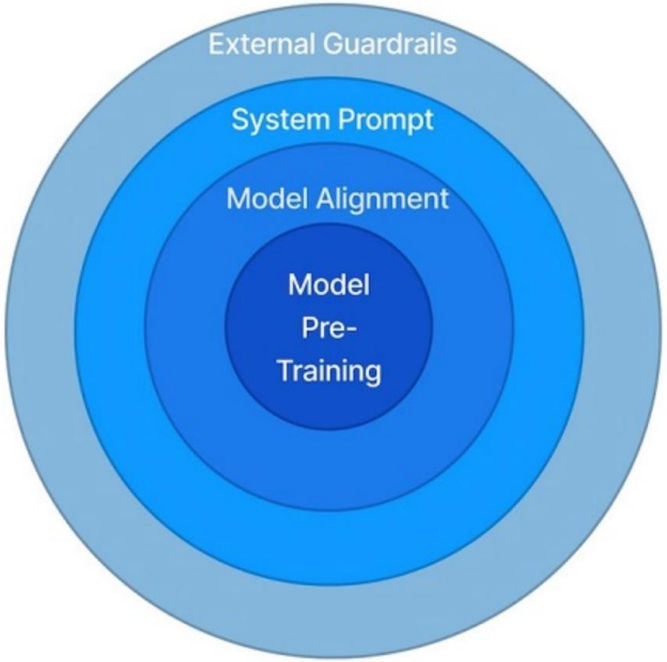
Safety Mechanisms for AI Application

Model producers (e.g., orgnizations) have the following responsibilities:

- **Data Preparation**: Data from various sources must be cleaned and organized before training a base model.
- **Value Alignment**: The model should be **aligned with values** such as accuracy, safety, and control using methods like **RLHF** or **DPO** after training.
- **Transparency**: Provide documentation or information (such as model cards) to explain the model's development process, limitions and intended use.

Model consumers (e.g., deployers) should also take certain actions:

- **Select a Base Model:** Choose a model that fits the specific use case and aligns with desired values.
- **Fine-Tune the Model:** Adjust the model to ensure it performs well in the target application.
- **Design Prompt Templates:** Create structured prompts to guide user inputs and outputs for better performance and safety.
- **Implement External Guardrails:** Add safety layers such as validation checks and filters to monitor and control inputs and outputs, ensuring secure and responsible use.



https://aws.amazon.com/blogs/machine-learning/build-safe-and-responsible-generative-ai-applications-with-guardrails/

Responsible Considerations for AI Application

1. Define the use case

A technology is not a use case. Face recognition is a technology, not a use case. The use case is the real-world scenario you apply the technology to, and it decides which errors you can afford.

Don't start with the technology, start with the user, their problem, and how you will address it.

Gallery retrieval

Confounding variation
Aging, makeup, hair



Possible bias
Race, age, gender

Consequences
Denied access to resources

Tuning
Favor recall or precision

Celebrity recognition

Confounding variation
Makeup, aging, pose, motion blur, occlusion, expression



Possible bias
Race, age, gender

Consequences
Missed sequence in media

Tuning
Favor precision

Virtual proctoring

Confounding variation
Background, pose, camera quality, occlusion



Possible bias
Race, age, gender, income

Consequences
False accusation

Tuning
Favor precision

https://skillbuilder.aws/learn/1H631ZWCTP/responsible-artificial-intelligence-practices/BN51NEFJNG

Responsible Considerations for AI Application

2. Model selection

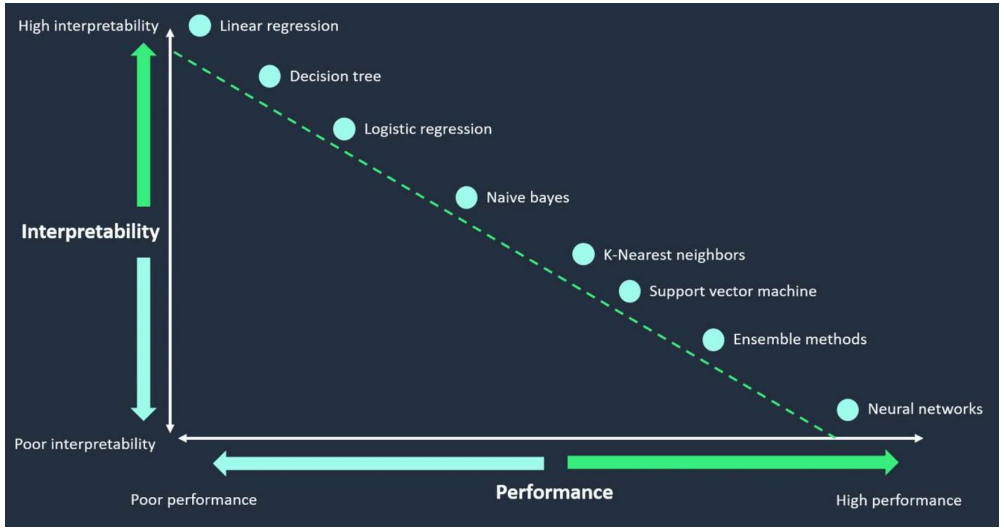
Model performance varies across several factors, including the following:

- **Model size** – More parameters mean higher capability but greater cost/environmental impact.
- **Level of customization** – The ability to change a model's output with new data ranging from prompt-based approaches to full model retraining
- **Inference options** – From self-managed deployment to API calls
- **Latency** – The amount of time it takes for a model to generate an output
- **Licensing agreements** – Some agreements can restrict or prohibit commercial use
- **Context windows** – The amount of information that can fit in a single prompt
- **Performance on datasets** - Model that performs differently on different datasets, and datasets are not constant, they are evolving.

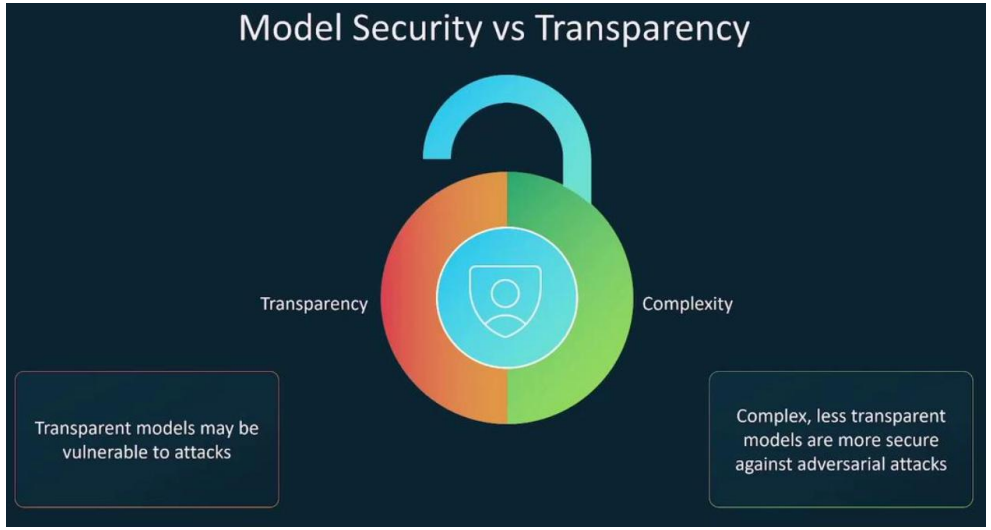
Responsible Considerations for AI Application

3. Model Trade-offs

- **Performance vs. transparency** — A simpler model (e.g., linear regression) is more transparent than a complex model (e.g., a neural network) but often sacrifices performance.
- **Security vs. transparency** — Balancing model security and transparency is equally critical. Revealing too much about a model's design or training data could result in vulnerability to attacks.
- **Capability vs cost** — more parameters mean higher capability and greater cost and environmental impact.



Performance vs. Transparency



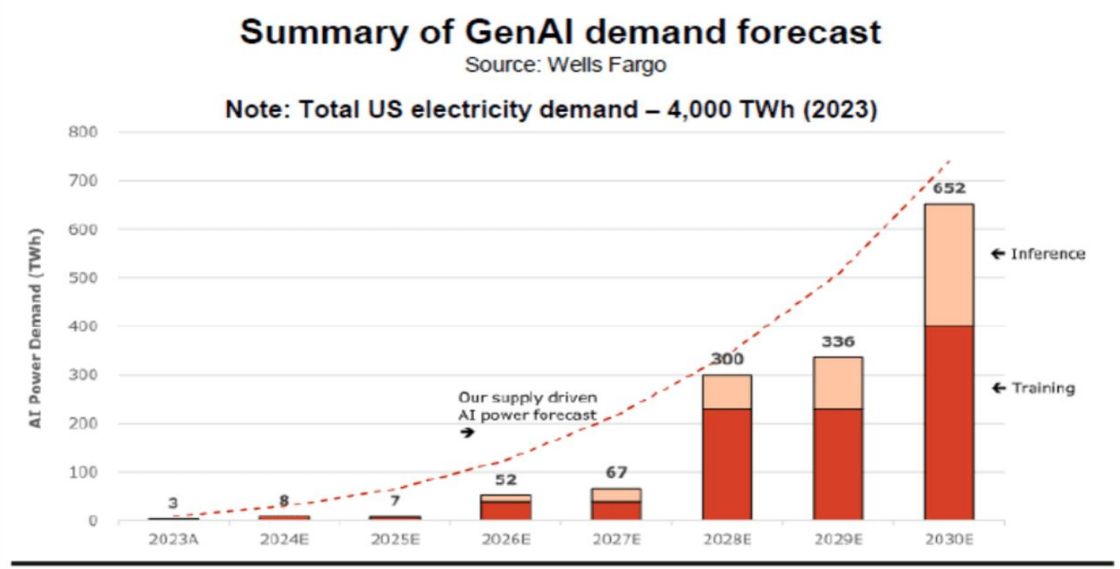
Security vs. Transparency

https://render.skillbuilder.aws/?module_id=R9WKSD3ST9%3A001.000.000&product_id=BN51NEFJNG%3A001.003.000&registration_id=3e744499-67ed-50a3-95b0-755b5691d9c0&navigation=digital

Responsible Considerations for AI Application

4. Economic and Enviromental Impacts

- **Environmental impact**: Training and running AI models at large-scale can consume significant amounts of energy.
- **Economic considerations:** Economic considerations in responsible AI include the potential benefits and costs of AI technologies and the impact on jobs and the economy.



	2023A	2024E	2025E	2026E	2027E	2028E	2029E	2030E
(+) Training power demand (TWh)	3	8	7	40	40	229	229	402
(+) Inference power demand (TWh)	0	0	0	12	27	70	107	250
Demand driven AI power forecast (TWh)	**3**	**8**	**7**	**52**	**67**	**300**	**336**	**652**
Supply driven AI power forecast (TWh)	*9*	*29*	*65*	*125*	*217*	*341*	*508*	*739*

Source: Wells Fargo Securities, LLC estimates

China vs. US Major Commercial LLM API Prices

Context: K tokens; Input / Output: M tokens. US $

	Model	Company	Context	Input	Output
1	Claude 3 Opus	Anthropic	200	15.00	75.00
2	GPT-4-turbo	OpenAI	128	10.00	30.00
3	GPT-4o	OpenAI	128	5.00	15.00
4	HunYuan-Pro	Tencent	32	4.13	13.77
5	Claude 3 Sonnet	Anthropic	200	3.00	15.00
6	HunYuan-Standard	Tencent	256	2.07	8.26
7	Doubao Pro	Bytedance	128	0.69	1.24
8	Qwen-Plus	Alibaba	-	0.55	1.65
9	Qwen-turbo	Alibaba	-	0.28	0.83
10	DeepSeek V2	Fantom	32	0.14	0.28
11	GLM-3-turbo	Zhipu AI	128	0.14	0.28
12	Doubao Pro	Bytedance	32	0.11	0.28
13	Qwen-Long	Alibaba	1000	0.07	0.28
14	ERNIE Speed	Baidu	8 / 128	0.00	0.00
15	ERNIE Lite	Baidu	8 / 128	0.00	0.00
16	HunYuan-Lite	Tencent	256	0.00	0.00

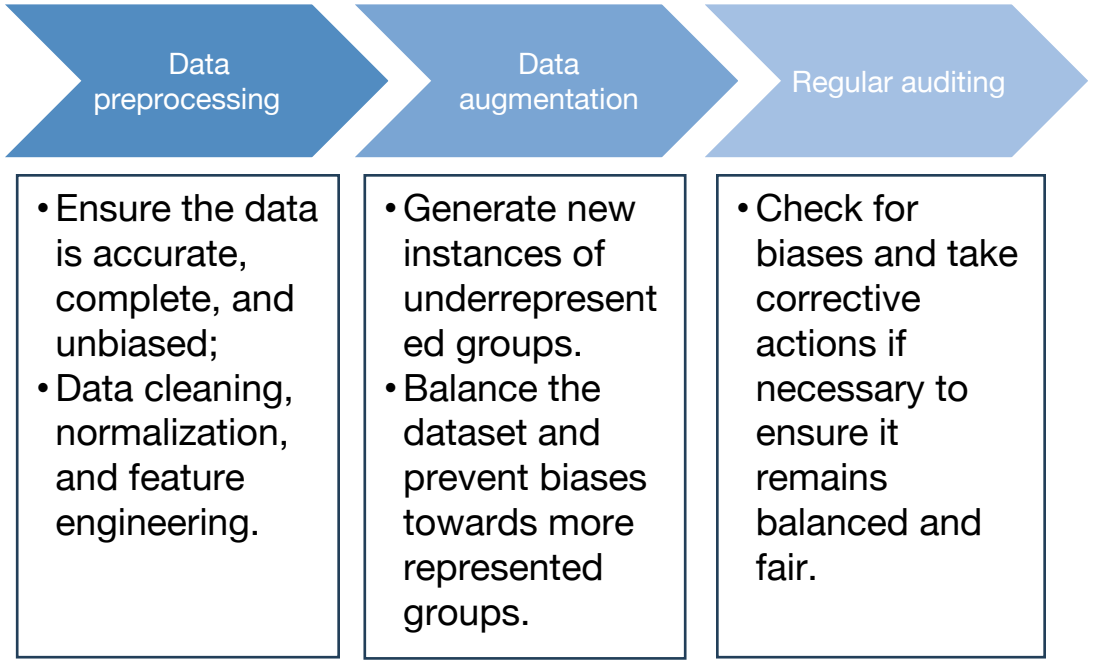
As of May 25

Table: @leeron • Created with Datawrapper

https://www.marketwatch.com/story/ai-could-demand-a-shocking-amount-of-electricity-check-out-this-chart-e91e306d
https://x.com/leeron/status/1794590078721040412

Responsible Preparation for Datasets

"*All want innovation but they don't have their data house in order.*" — Prof. Dean Mohamedally, UCL Industry Exchange Network



Before any of that — check data from the following five aspects:

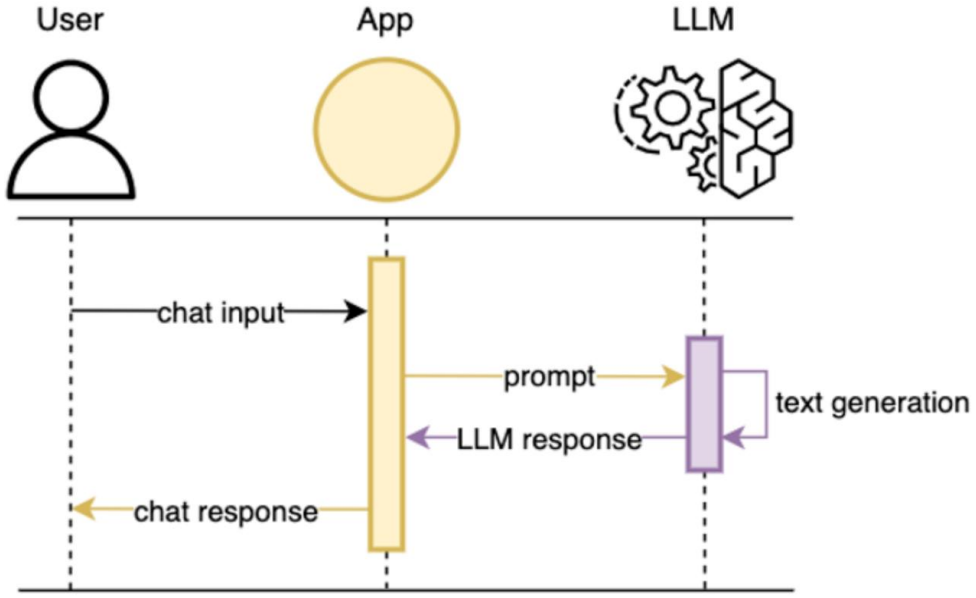
- **Data exists** — the dataset required for the intended task is actually held by the organisation.
- **Data is clean** — a critical mass of records is usable, or a viable path to that state exists.
- **Data is documented** — the meaning of every field is recorded and understood.
- **Data is accessible** — records are retrievable rather than confined to paper files or legacy systems.
- **Ownership is assigned** — a named individual holds authority over data-access decisions.

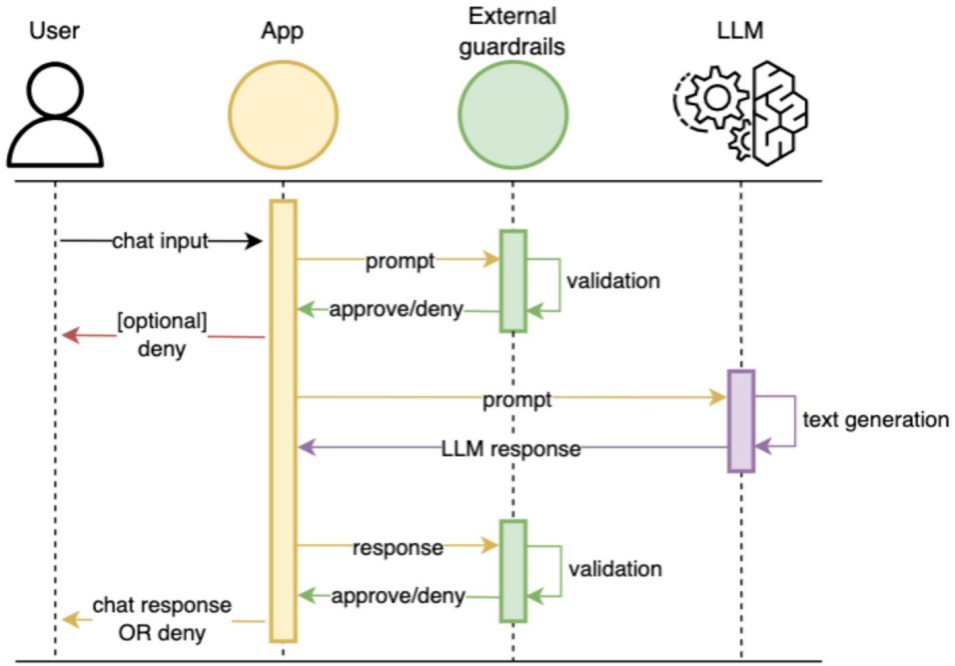
https://skillbuilder.aws/learn/1H631ZWCTP/responsible-artificial-intelligence-practices/BN51NEFJNG

Implement External Guardrails

Adding External Guardrails to LLM Applications

- **Standard flow:** chat message -> text generation -> chat response
- **Flow with guardrails:** chat message -> input validation -> text generation -> output validation -> response





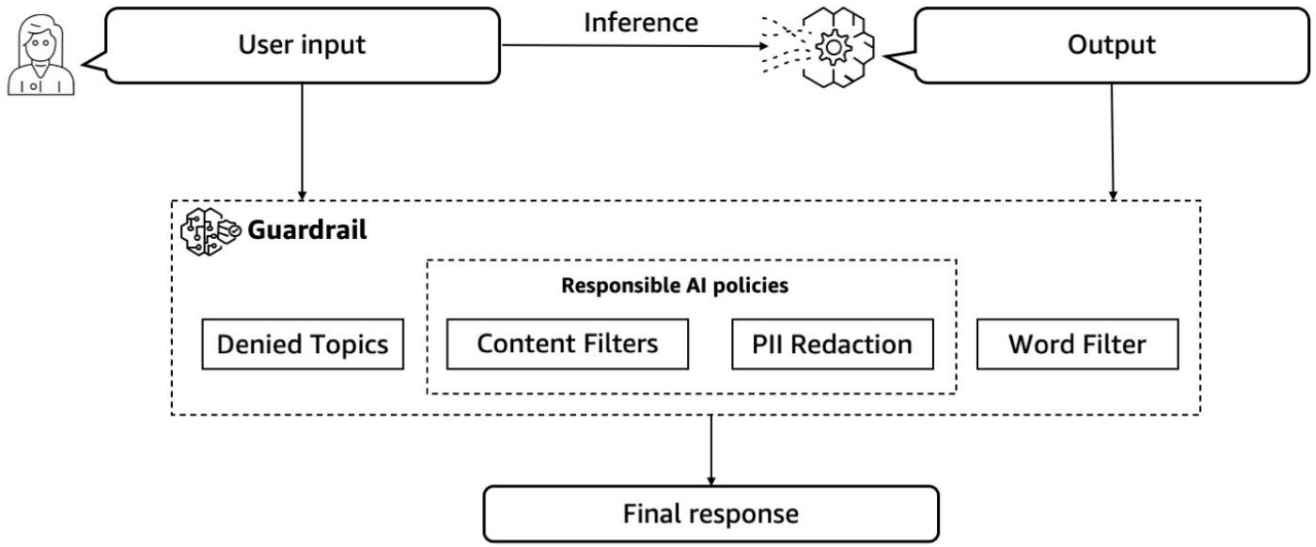
https://aws.amazon.com/blogs/machine-learning/build-safe-and-responsible-generative-ai-applications-with-guardrails/

Implement External Guardrails

Case Study: Amazon Bedrock Guardrails

Amazon Bedrock Guardrails consists of a collection of **filtering policies** that users can configure to avoid undesirable and harmful content and remove or mask sensitive information for privacy protection:

- **Word filter**: Configure custom dictionaries to detect and block specific words or phrases—such as profanity, competitor names, or other offensive terms.
- **Content filter**: Set thresholds to block harmful prompts and responses, e.g., hate speech, violent content, and malicious prompt attacks.
- Others: denied topics, sensitive information filters, Intent classification, toxicity detection ...



https://aws.amazon.com/blogs/machine-learning/build-safe-and-responsible-generative-ai-applications-with-guardrails/

Evaluating an LLM Application

Red Teaming and Incident Data

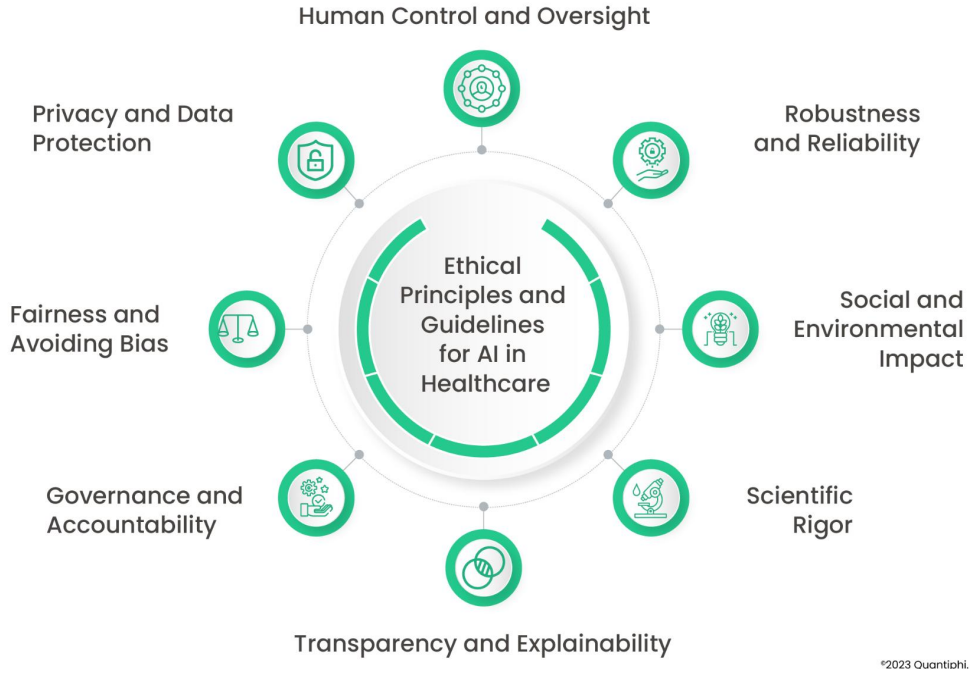
Guardrails (previous two slides) block *known-bad* inputs and outputs. They do not tell you whether *your* app is safe. For that, you have to test it.

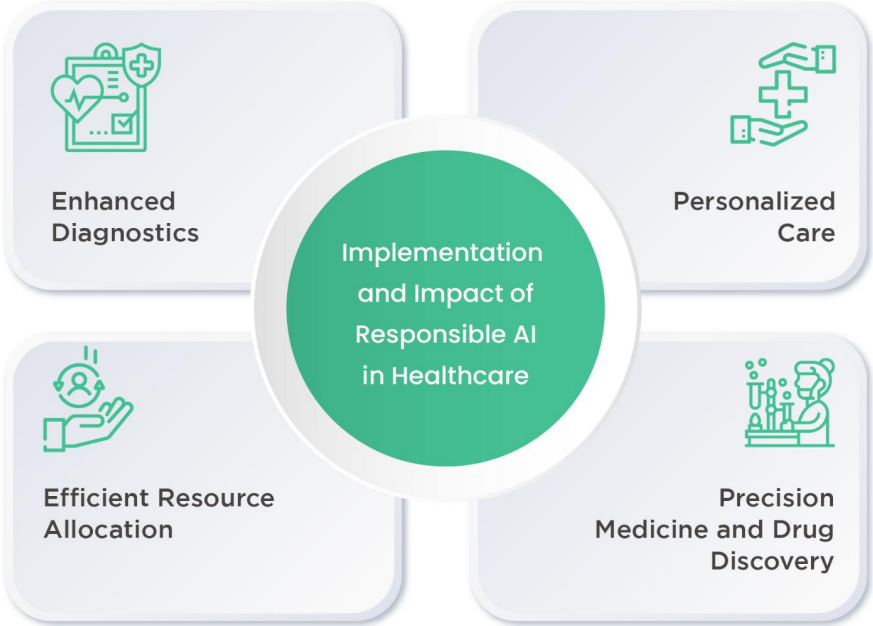
- **Model cards** — a short document stating intended use, out-of-scope use, evaluation data and known limitations (Mitchell et al., 2019). Ask a vendor for one. **If there is none, that is itself a risk signal.**
- **Red teaming** — your team deliberately attacks your own application: prompt injection, jailbreak, extracting the system prompt, asking for PII, asking questions outside the knowledge base. Every successful attack becomes a permanent test case.
- **Use an existing checklist** so you don't rediscover known problems. OWASP Top 10 for LLM Applications (2025)
- **Use incident data instead of imagination** — the AI Incident Database (incidentdatabase.ai) and the OECD AI Incidents Monitor collect real-world AI failures, searchable by harm type. Check what has already gone wrong in your domain before you ship.

Mitchell et al., "Model Cards for Model Reporting" (2019) · OWASP Top 10 for LLM Applications 2025 · incidentdatabase.ai · oecd.ai/en/incidents

Case Study: Responsible AI in Healthcare

- **Diagnostic Tools:** AI models assist in disease identification, reducing diagnostic biases and enhancing accuracy.
- **Treatment Recommendation Systems:** Ethical AI frameworks guide treatment plans, preventing discriminatory practices.
- **Predictive Analytics for Patient Care**: Responsible AI analyzes patient data to predict potential health risks, enabling early interventions while ensuring data privacy and fairness in decision-making.





https://quantiphi.com/blog/responsible-ai-in-healthcare-empowering-health-with-ethics-and-efficiency/

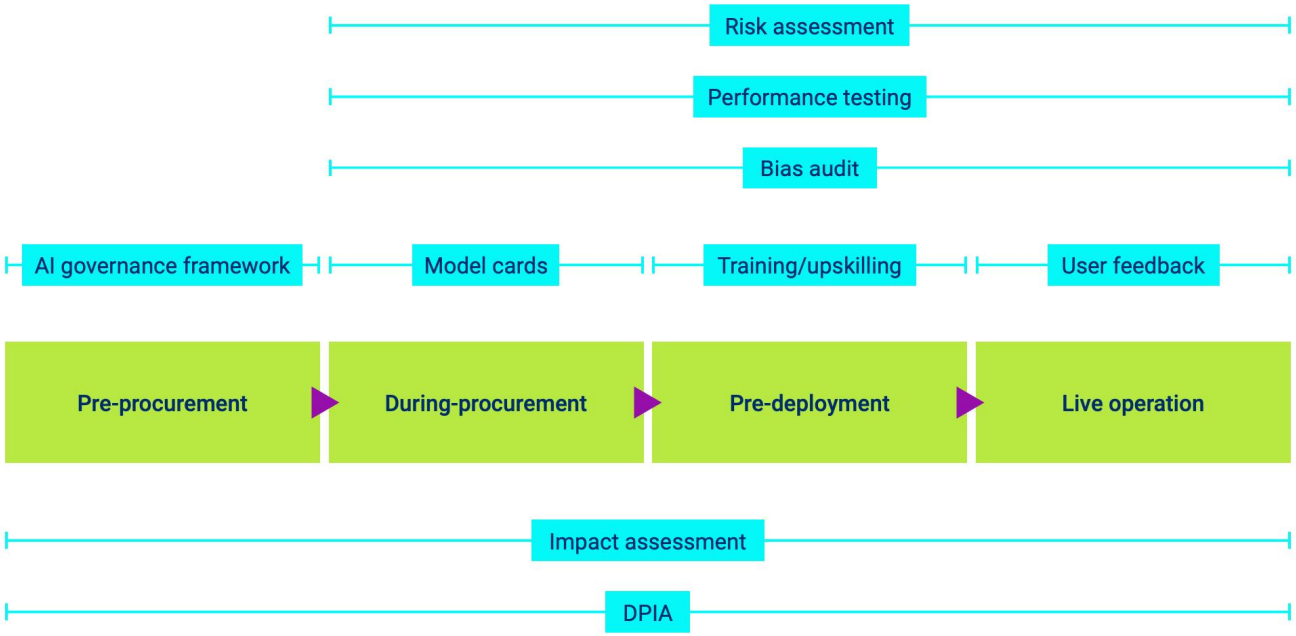
Case Study: Responsible AI in Hiring and HR

In the HR and recruitment sector, organizations are using a range of AI-enabled technologies across every stage of the recruitment process, including sourcing, screening, interview and selection

- **Pre-procurement:** Planning and identifying AI needs, setting requirements, and defining governance structures.
- **During-procurement:** Selecting vendors, assessing models, and ensuring compliance with standards.
- **Pre-deployment:** Testing and validating models before launch.
- **Live operation**: Ongoing monitoring, evaluation, and improvement after deployment.



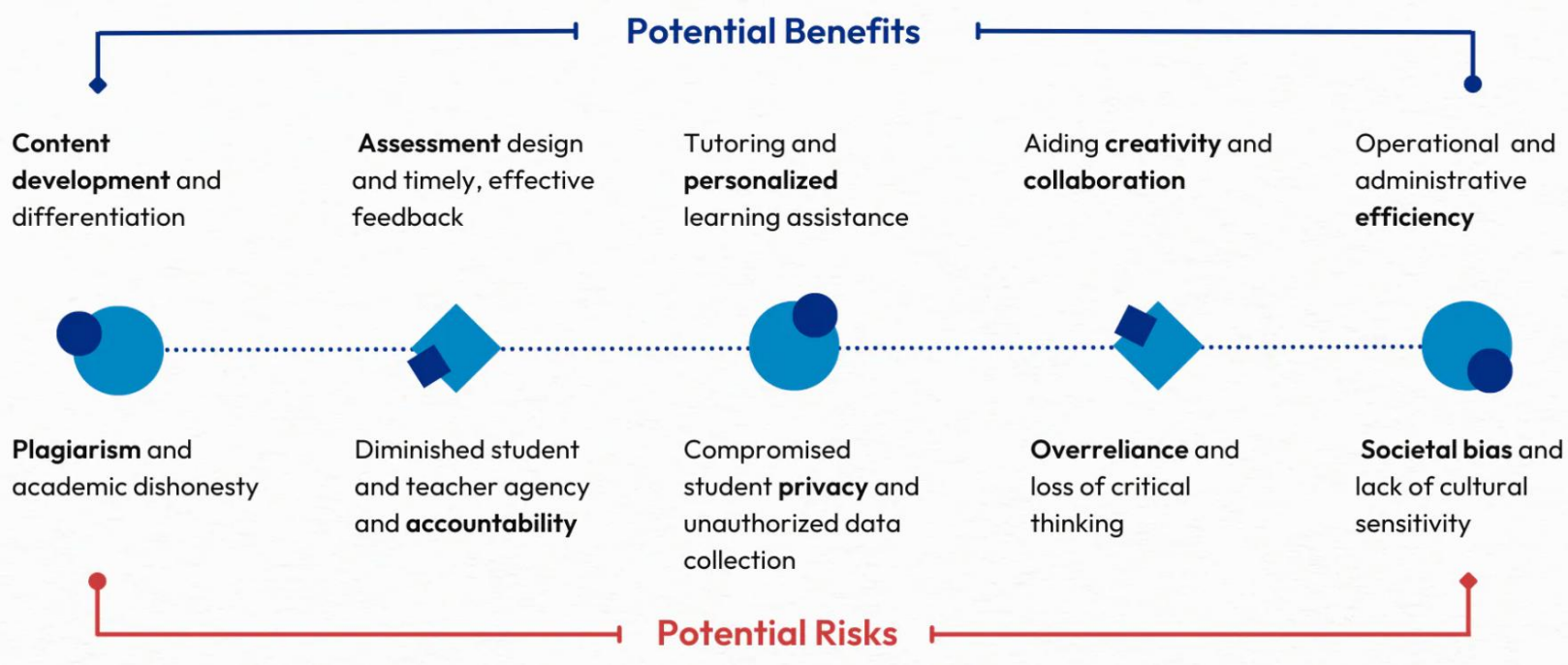
Assurance mechanisms for AI in recruitment



https://www.gov.uk/government/publications/responsible-ai-in-recruitment-guide/responsible-ai-in-recruitment
https://assets.publishing.service.gov.uk/media/65fda1b9f1d3a0001132ae5b/Responsible_AI_in_Recruitment.pd

Case Study: Responsible AI in Education

- **Adaptive Learning Platforms**: AI customizes educational content to meet diverse student needs.
- **Performance Monitoring**: AI tracks student progress, providing insights.
- **Assessment Tools**: AI grades assessments impartially, ensuring **fairness** in academic evaluations.
- **Essay Writing Tools**: Students can use responsible AI tools to improve academic writing skill with AI, ensuring their work meets academic standards while **maintaining ethical integrity**.

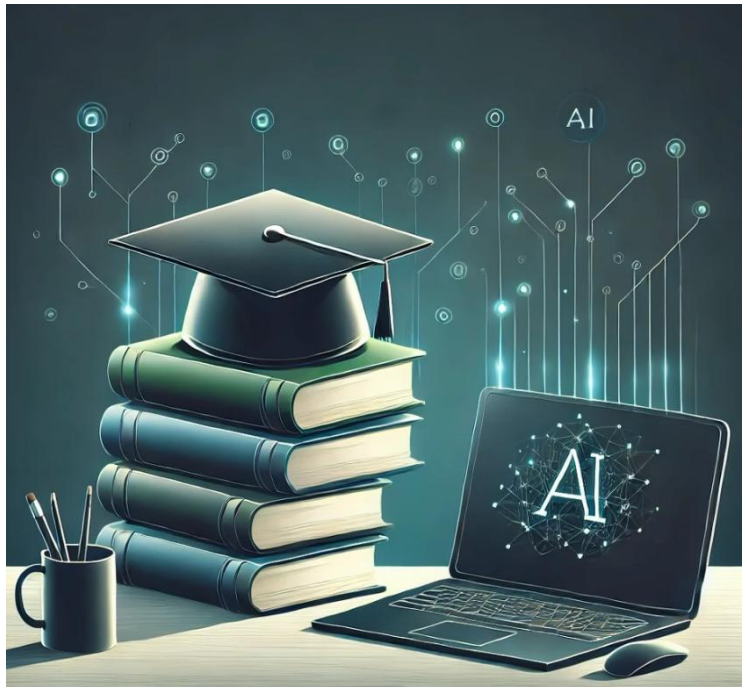


https://www.weforum.org/stories/2024/01/ai-guidance-school-responsible-use-in-education/

Using AI in Your Own Work

Questions for Critically Evaluating AI or AI-generated Content in Academics

- Is the AI-generated content accurate? How can you test or assess the accuracy?
- Can other credible sources (**GenAI is not credible source**) validate the data or item produced?
- How does the information generated impact or influence your thinking on this topic?
- Who is represented in this data? Is the data inclusive in terms of the material's scope and the perspectives that it presents?
- Knowing LLMs may also be collecting the data your students input (i.e., in their prompts), how will you safeguard your own privacy?



https://teaching.cornell.edu/generative-artificial-intelligence/ethical-ai-teaching-and-learning

Summary

- **AI safety is broader than accidents**

  Risks from AI are concrete: bias, misinformation, deepfakes, and the MIT taxonomy sorts them by cause, domain and timing. The documented failures show why: biased screening, opaque welfare scoring, a chatbot that invented a policy.

- **Responsible AI names the principles**

  Fairness, explainability, robustness, transparency and privacy describe a trustworthy system. Codes of conduct, governance frameworks and review boards are what make them enforceable rather than aspirational.

- **Building responsibly is a sequence of decisions**

  Define the use case, select the model, prepare the data, add guardrails, then evaluate the result. Healthcare, hiring and education show the same sequence facing different risks.